

Lab C — Microphone calibration

34870 Electroacoustics

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Reference version. This is a complete worked version of the lab, written to check the group's own report against: the numbers, figures and conclusions below are what the data give. The group's report is `LabC_report.tex`. All numbers come from `LabC/matlab/run_all.m`, which reads the untouched lab-PC files in `LabC/rawdata/` and writes the figures in `LabC/figures/`.

1 Purpose and set-up

Two ½-inch condenser microphones, a B&K 4133 (free field) and a B&K 4134 (pressure field), were calibrated in two steps. First the absolute sensitivity was measured at a single frequency with three GRAS 42AG sound calibrators. Then the frequency response was measured with an electrostatic actuator, which drives the diaphragm with a force that is practically independent of frequency. The response gives the shape, and the calibrator gives the level. Resonance frequency and Q were read off the response and used to set the backplate parameters of an LTspice model of the microphone.

All measurements used the NI USB-4431 card (96 kHz sampling) and the B&K Nexus amplifier set to 10 mV/Pa in and out ($G_{\text{Nex}} = 1$), 0.1 Hz/100 kHz filters and 200 V polarisation. The capsules were only labelled “Mic 1” and “Mic 2” in the lab; which type is which follows from the measured responses (Section 4.2). Serial numbers, calibrator certificates and room conditions were not recorded; the room was at ordinary laboratory conditions.

2 Part 1: the measurement system

The generator output AO0 was connected to input AI0 and, through the Nexus, to input AI1. The ratio

$$H_{21,\text{ref}}(f) = \frac{V_{\text{AI1}}(f)}{V_{\text{AI0}}(f)} \quad (1)$$

is the response of the measurement chain alone. It is divided out of every later measurement: at 250 Hz and 1 kHz in Part 2, and at every frequency in Part 3.

The multitone has six lines per octave from 20 Hz up to the requested 60 kHz, but the card cannot measure anything above the Nyquist frequency $f_s/2 = 48$ kHz. The line at 47.8 kHz lies on the anti-aliasing edge ($|H_{21,\text{ref}}| = -5.8$ dB), and the line at 53.6 kHz is a mirror image of 42.4 kHz. Both are left out, so the usable band is 20 Hz to 42.7 kHz (68 lines).

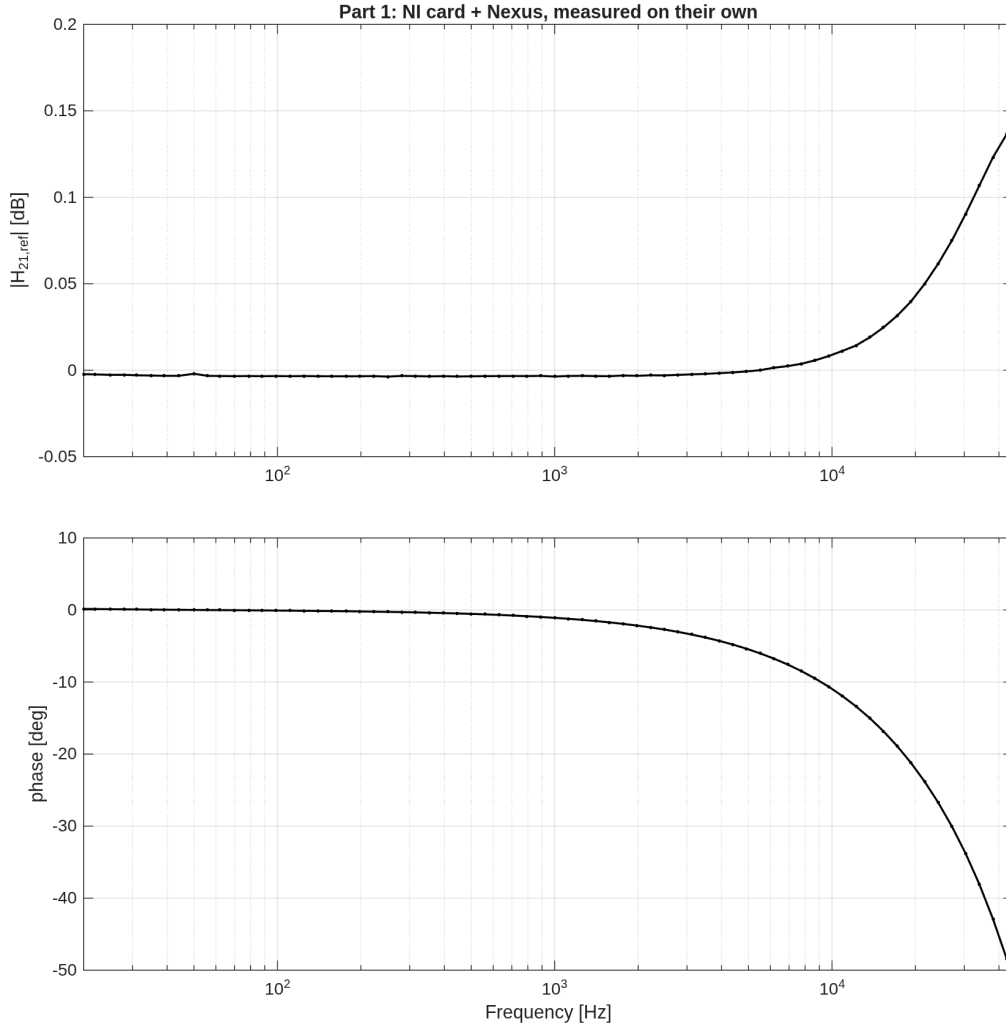


Figure 1: System reference $H_{21,\text{ref}}$. The magnitude stays within 0.15 dB of 0 dB. The phase falls linearly to -48° at 42 kHz, which is a time delay of 3.2 μs between the two input channels.

The phase is the reason the correction matters. Without it, the resonance frequency read from the -90° crossing would come out several kHz too low. At the two calibration frequencies

$$|H_{21,\text{ref}}(250 \text{ Hz})| = 0.99957 \text{ } (-0.0037 \text{ dB}), \quad |H_{21,\text{ref}}(1 \text{ kHz})| = 0.99958 \text{ } (-0.0037 \text{ dB}). \quad (2)$$

3 Part 2: sensitivity at one frequency

Each microphone was placed in each of the three calibrators, at 250 Hz and at 1 kHz, with the frequency and level from each calibrator's label. The course function `Calibrate` returns

$$M = \frac{v_{\text{rms}}}{p_{\text{rms}} G_{\text{Nex}}}, \quad p_{\text{rms}} = 20 \text{ } \mu\text{Pa} \cdot 10^{L_p/20}, \quad (3)$$

which is then divided by $|H_{21,\text{ref}}|$ at the same frequency.

Correction of the stored values. On the lab PC the *dB values* of $H_{21,\text{ref}}$ (-0.003607 and -0.003672) were typed in instead of the linear magnitudes (0.99957 and 0.99958). The stored sensitivities were therefore divided by 0.0036 instead of by 1, and they read about 3 V/Pa. The correction is

$$M = M_{\text{stored}} \cdot \frac{|\text{typed value}|}{|H_{21,\text{ref}}|}. \quad (4)$$

Table 1 shows the corrected values.

Table 1: Corrected sensitivities in mV/Pa.

Calibrator	Mic 1 (4134)		Mic 2 (4133)	
	250 Hz	1 kHz	250 Hz	1 kHz
42AG no. 1	10.530	11.003	12.547	13.059
42AG no. 2	10.531	10.960	12.559	13.010
42AG no. 3	10.561	10.970	12.579	13.009
Mean	10.541	10.978	12.562	13.026
Std. deviation	0.017	0.023	0.016	0.028
Relative std. dev.	0.16 %	0.21 %	0.13 %	0.22 %
Level re 1 V/Pa	−39.54 dB	−39.19 dB	−38.02 dB	−37.70 dB

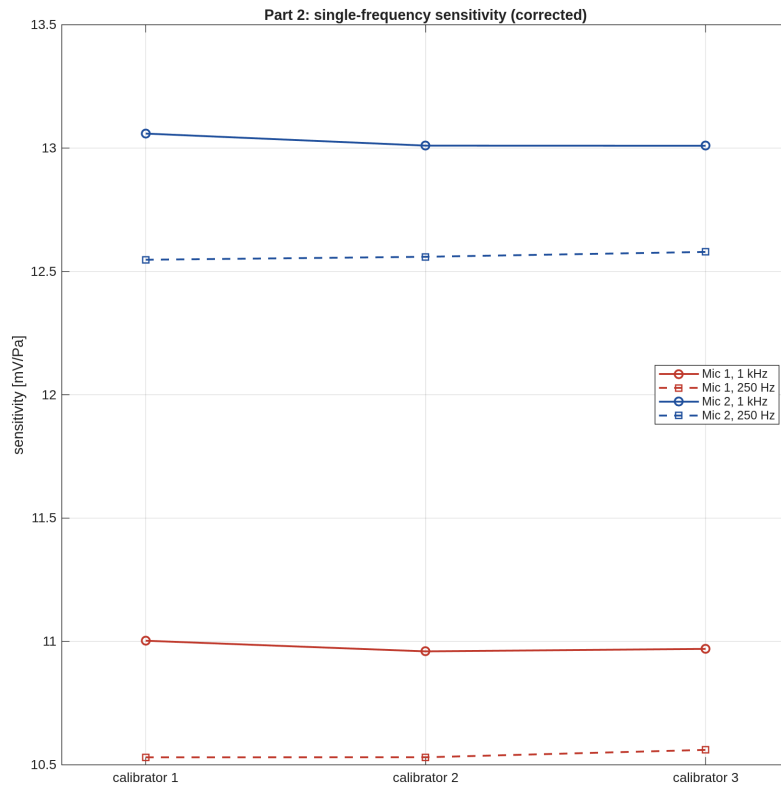


Figure 2: Sensitivity for each calibrator. The spread between calibrators is small, but the 250 Hz results sit about 0.35 dB below the 1 kHz results for both microphones.

Repeatability, reproducibility and true value.

- **Repeatability.** With the same microphone, the same method and three calibrators, the results agree within 0.2 % (0.02 dB). The measurement itself is very stable.
- **Reproducibility.** Changing the method, here from 1 kHz to 250 Hz, moves the result by −0.35 dB for Mic 1 and −0.32 dB for Mic 2, about 4 %. Both microphones are flat within 0.1 dB between the two frequencies (Part 3), so the difference is not in the microphones. It is the same for all three calibrators, so it is systematic. The most likely cause is `Calibrate.m`: at 250 Hz it sums only six FFT bins ending at the nominal frequency (the window was written for a 251.2 Hz pistonphone), while at 1 kHz it sums seventeen bins around the tone. A tone slightly above 250 Hz loses part of its power outside the window. The time signals were not saved, so this is the most plausible explanation, not a proven one.
- **True value.** The small spread shows that the method is repeatable, not that it is right: a systematic error of 4 % is ten times larger than the spread and does not show up in it. How close the result is to the true value depends on the traceable uncertainty of the calibrators, not on the scatter of the readings.

The 1 kHz values are used from here on: Mic 1 $M = 10.98$ mV/Pa, Mic 2 $M = 13.03$ mV/Pa.

Uncertainty budget (1 kHz, Mic 1). The certificates were not recorded. The calibrator figure below is therefore the class-1 tolerance of IEC 60942, taken as an assumed value, not a certificate value.

Contribution	Basis	u [dB]	u [%]
Calibrator level	± 0.2 dB tolerance (assumed), rectangular: $0.2/\sqrt{3}$	0.115	1.3
Repeatability of the mean	$s/\sqrt{3} = 0.023/\sqrt{3}$ mV/Pa	0.011	0.12
System correction $ H_{21,\text{ref}} $	measured, varies by < 0.001 dB between lines	0.001	0.01
Static pressure / temperature	not measured, ordinary room (assumed ± 0.05 dB, rectangular)	0.029	0.33
Combined u_c	root sum of squares	0.12	1.4
Expanded U ($k = 2$, about 95 %)		0.24	2.8

Result: Mic 1 (4134) $M = 10.98 \text{ mV/Pa} \pm 0.31 \text{ mV/Pa}$, Mic 2 (4133) $M = 13.03 \text{ mV/Pa} \pm 0.36 \text{ mV/Pa}$ ($k = 2$). The budget is dominated by the calibrator. The 0.35 dB method offset at 250 Hz is larger than the whole budget. That is why a known systematic effect has to be removed or explained, not simply added to the spread.

4 Part 3: frequency response with the actuator

4.1 Noise floor

With the actuator amplifier switched off, the microphone output was recorded with 4 and with 64 averages (Mic 2 only). The noise is shown in the same normalisation as the response (microphone channel over drive channel).

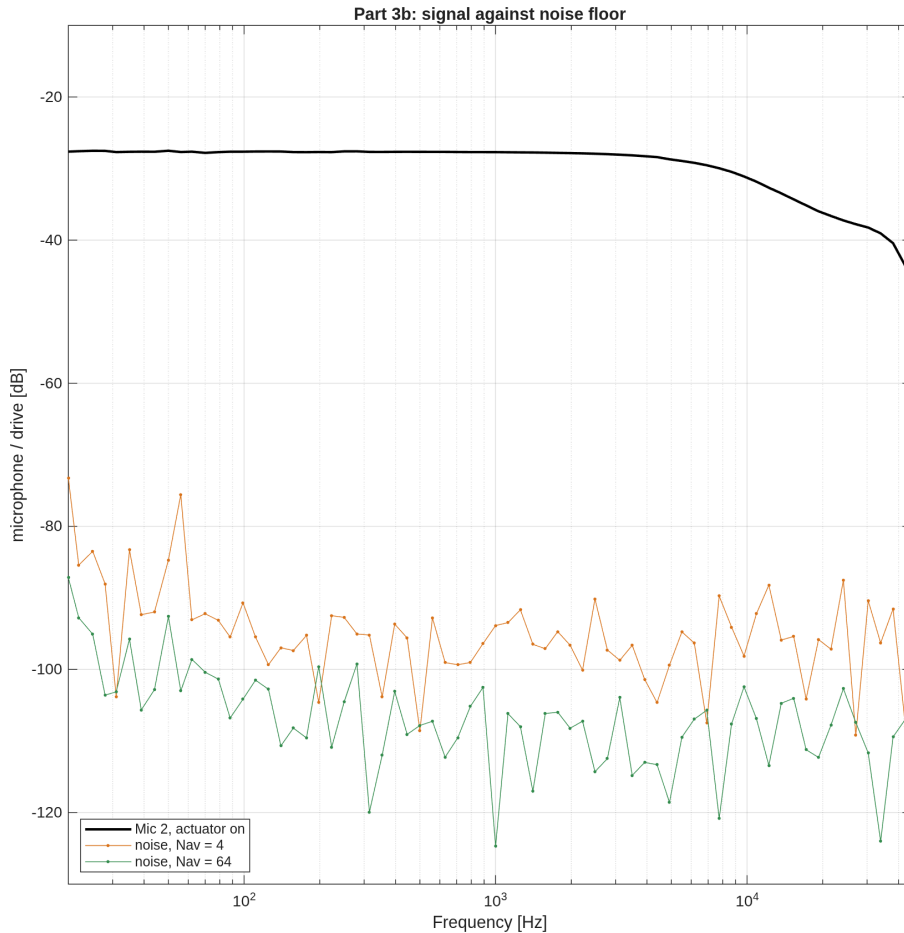


Figure 3: Actuator response of Mic 2 against the noise floor.

The signal is 60 to 90 dB above the noise over most of the band. The worst case is 46 dB (4 averages) and 60 dB (64 averages), both at 20 Hz, where the room noise is highest. Going from 4 to 64 averages lowered the noise by a median of 11.6 dB. Averaging uncorrelated noise predicts $10 \log_{10} 16 = 12.0$ dB, so the noise is random and not a disturbance locked to the signal. The noise has no influence on the responses.

4.2 Responses and identification of the microphones

The measurement system is removed by

$$H_{pv}(f) = \frac{H_{21}(f)}{H_{21,\text{ref}}(f)}, \quad (5)$$

and each curve is scaled to its 1 kHz sensitivity,

$$M(f) = M(1 \text{ kHz}) \frac{H_{pv}(f)}{|H_{pv}(1 \text{ kHz})|}. \quad (6)$$

The calibrator is a small closed coupler, so like the actuator it gives the *pressure* sensitivity. At 1 kHz the free-field correction of a ½-inch microphone is practically zero, so the 1 kHz value is the right one for both microphones. It is also the calibration without the systematic offset.

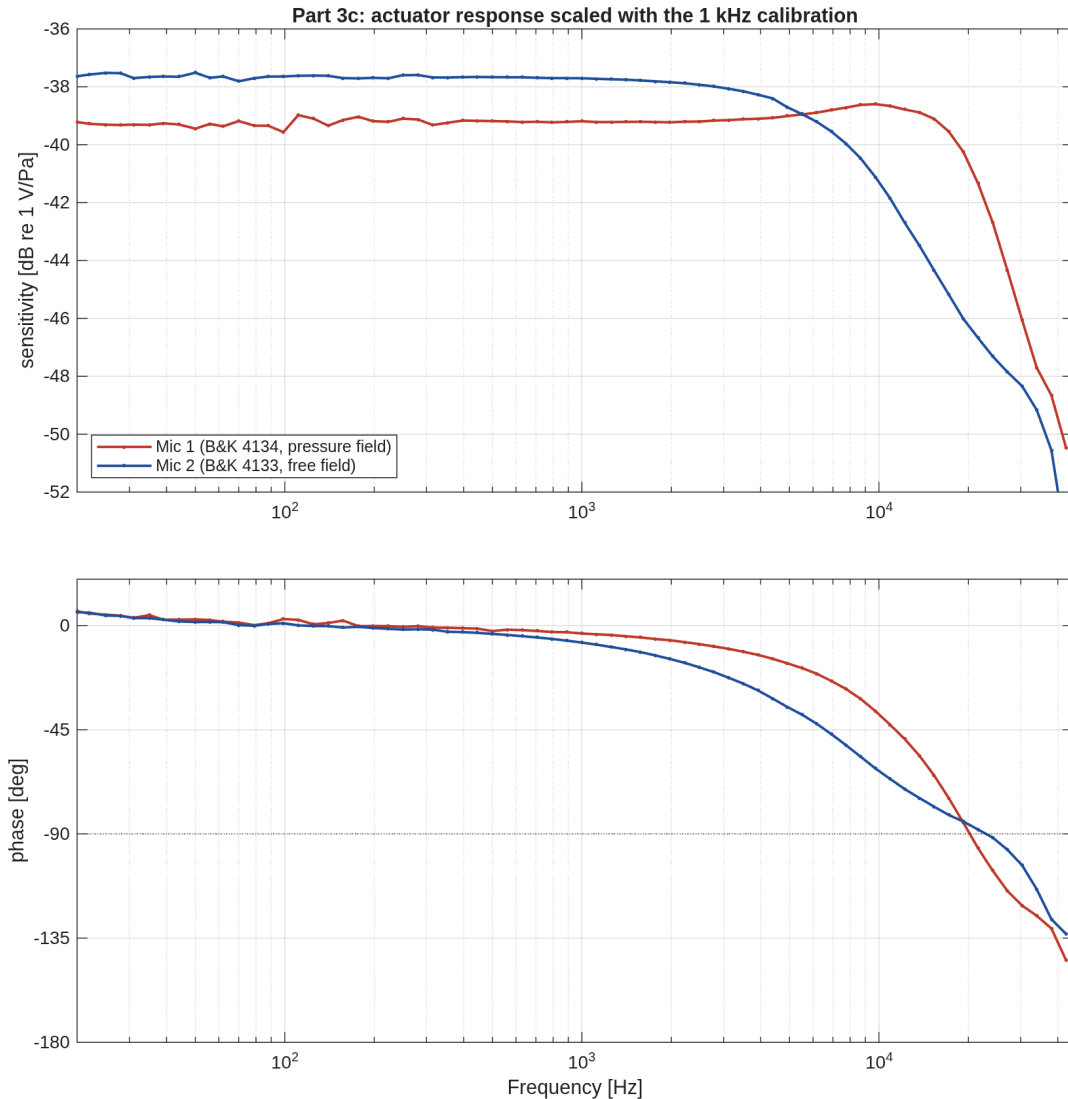


Figure 4: Actuator responses scaled with the 1 kHz calibration.

- **Mic 1** is flat, rises by 0.55 dB around 10 kHz and is 1.4 dB down at 20 kHz. A flat response to a uniform pressure is the design goal of a *pressure-field* microphone, so Mic 1 is the **B&K 4134**.

- **Mic 2** is 3.6 dB down at 10 kHz and 8.6 dB down at 20 kHz. The diaphragm is deliberately over-damped: in a free field, diffraction around the microphone raises the pressure on the diaphragm at high frequencies (the same scattering measured on the cylinder in Lab B). The falling pressure response cancels that rise. Mic 2 is therefore the *free-field* microphone, the **B&K 4133**.

5 Resonance frequency and Q

Following the course slides (lectures 4 and 6), below resonance the diaphragm is stiffness controlled (phase 0°), above it mass controlled (-180°). The resonance is where

$$\angle H_{pv}(f_s) = -90^\circ, \quad Q = \frac{|H_{pv}(f_s)|}{|H_{pv}(f \ll f_s)|}, \quad (7)$$

with log-frequency interpolation between the measured lines and the low-frequency level taken as the mean over 100 Hz–500 Hz. As a cross-check, the second-order function

$$H(f) = \frac{G}{1 - (f/f_s)^2 + j(f/f_s)/Q} \quad (8)$$

was fitted by least squares to the complex H_{pv} from 100 Hz to 42.7 kHz.

Microphone	Slide procedure		Least-squares fit	
	f_s [kHz]	Q	f_s [kHz]	Q
Mic 1, 4134 (pressure)	20.22	0.835	20.78	0.813
Mic 2, 4133 (free field)	22.93	0.340	21.08	0.388

For the 4134 the two methods agree within 3 %. For the 4133 they differ by 9 % in f_s and 14 % in Q . The reason is visible in the phase: it falls slowly between about 12 and 25 kHz and then faster than a single mass–spring–damper would. The strong damping of a free-field capsule comes from the air film between the diaphragm and a slotted, perforated backplate, and that damping depends on frequency. A one-point reading and a whole-curve fit then weight different parts of the curve.

6 Model of the condenser microphone

6.1 Parameters

The brief gives the same data for both capsules: gap $x_0 = 20.77 \text{ }\mu\text{m}$, polarisation $E = 200 \text{ V}$, effective diameter 8.95 mm, diaphragm mass $M_{MD} = 1.5 \text{ mg}$, compliance $C_{MD} = 0.02 \text{ mm/N}$, no diaphragm damping, back volume $V_B = 126.4 \text{ mm}^3$. With $\rho = 1.18 \text{ kg/m}^3$ and $c = 344 \text{ m/s}$:

$$S_D = \pi a^2 = 6.291 \times 10^{-5} \text{ m}^2, \quad C_{AB} = \frac{V_B}{\rho c^2} = 9.05 \times 10^{-13} \text{ m}^5/\text{N}, \quad M_{A1} = \frac{0.6133\rho}{\pi a} = 51.5 \text{ kg/m}^4, \quad (9)$$

$$\frac{1}{C_{MT}} = \frac{1}{C_{MD}} + \frac{S_D^2}{C_{AB}} \Rightarrow C_{MT} = 18.4 \text{ }\mu\text{m/N}, \quad (10)$$

$$C_{E0} = \frac{\varepsilon_0 S_D}{x_0} = 26.8 \text{ pF}, \quad M_{\text{model}} = \frac{E C_{MT} S_D}{x_0} = 11.14 \text{ mV/Pa}. \quad (11)$$

The back volume stiffens the diaphragm by about 9 %. The measured f_s fixes the total moving mass and Q fixes the resistance, so the backplate mass and resistance follow:

$$M_{MT} = \frac{1}{(2\pi f_s)^2 C_{MT}}, \quad M_{AS} = \frac{M_{MT} - M_{MD}}{S_D^2} - M_{A1}, \quad R_{AS} = \frac{1}{Q S_D^2} \sqrt{\frac{M_{MT}}{C_{MT}}}. \quad (12)$$

Microphone	f_s [Hz]	Q	M_{AS} [kg/m ⁴]	R_{AS} [Pa s/m ³]
4134 (Mic 1)	20 221	0.835	420.5	1.29×10^8
4133 (Mic 2)	22 930	0.340	231.4	2.80×10^8

Both values are positive and physically sensible. The diaphragm alone would resonate at 28.4 kHz; the air moved through the backplate holes lowers this to the measured values. The free-field capsule has about twice the backplate resistance, which is the damping that gives it its low Q .

6.2 LTspice circuit

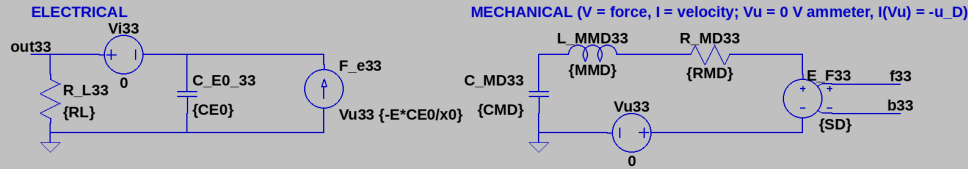
The circuit is drawn as in the lectures: an electrical loop and a mechanical loop side by side, with the acoustic network below. Mechanical and acoustic parts both use the impedance analogy (voltage = force or pressure, current = velocity or volume velocity). The domains are linked by controlled sources driven by a 0 V ammeter in the mechanical loop, whose current is the diaphragm velocity u_D : the force $S_D(p_{\text{front}} - p_{\text{back}})$, the volume velocity $S_D u_D$ and the electrical output current $(EC_{E0}/x_0) u_D$. The actuator is a 1 Pa source applied directly to the diaphragm. There is no G_{pb} generator for the diffraction function $T(s)$, because the actuator does not create a sound field. Only f_s and Q are typed in; LTspice computes M_{AS} and R_{AS} from them with the formulas above.

Lab C - condenser microphones 4133 / 4134 driven by an electrostatic actuator (pressure applied straight to the diaphragm: no sound field, no T(s) / Gpb generator) Layout as in the lectures: electrical loop | mechanical loop, acoustic network below. Only fs and Q are typed in; M_AS and R_AS follow inside LTspice.

```
.param a=0.004475 x0=2.077e-05 E=200 MMD=1.5e-06 CMD=2e-05 RMD=1e-9 Vb=1.264e-07 RL=1e+10 rho=1.18 c=344.0
.param SD=pi*a**2 CAB=Vb/(rho*c**2) MA1=0.6133*rho/(pi*a) CMT=1/(1/CMD+SD**2/CAB) CE0=8.854e-12*SD/x0
.ac dec 200 20 60k
```

B&K 4133, free field (Mic 2 in the lab) <- measured values:

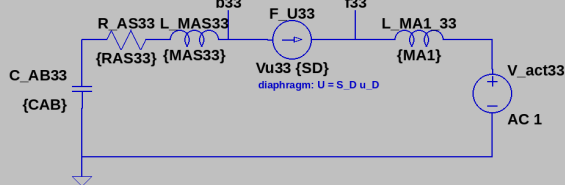
```
.param fs33=22930 Q33=0.34
.param MMT33=1/((2*pi*fs33)**2*CMT) MAS33=(MMT33-MMD)/SD**2-MA1 RAS33=sqrt(MMT33/CMT)/Q33/SD**2
```



ACOUSTIC (V = pressure, I = volume velocity)

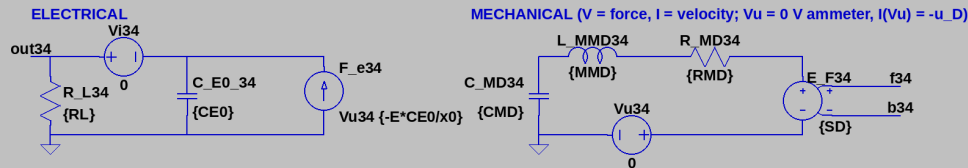
back: C_AB, R_AS, M_AS

front: M_A1 + actuator (1 Pa)



B&K 4134, pressure field (Mic 1 in the lab) <- measured values:

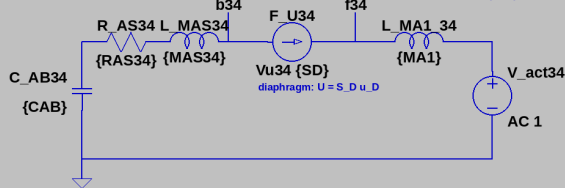
```
.param fs34=20221 Q34=0.835
.param MMT34=1/((2*pi*fs34)**2*CMT) MAS34=(MMT34-MMD)/SD**2-MA1 RAS34=sqrt(MMT34/CMT)/Q34/SD**2
```



ACOUSTIC (V = pressure, I = volume velocity)

back: C_AB, R_AS, M_AS

front: M_A1 + actuator (1 Pa)



plot V(out33) V(out34): dB re 1 V/Pa and phase. Low-frequency level E*SD*CMT/x0 = 11.1 mV/Pa (-39.1 dB), below the nominal 12.5 mV/Pa: see the brief's appendix. fs = where the phase is -90 deg, Q = |H(fs)| / |H(f << fs)| (linear). R_MD = 0 in the brief (1 nOhm here).

Figure 5: LTspice schematic LabC_CondenserMics.asc (rendered preview).

6.3 Model against measurement

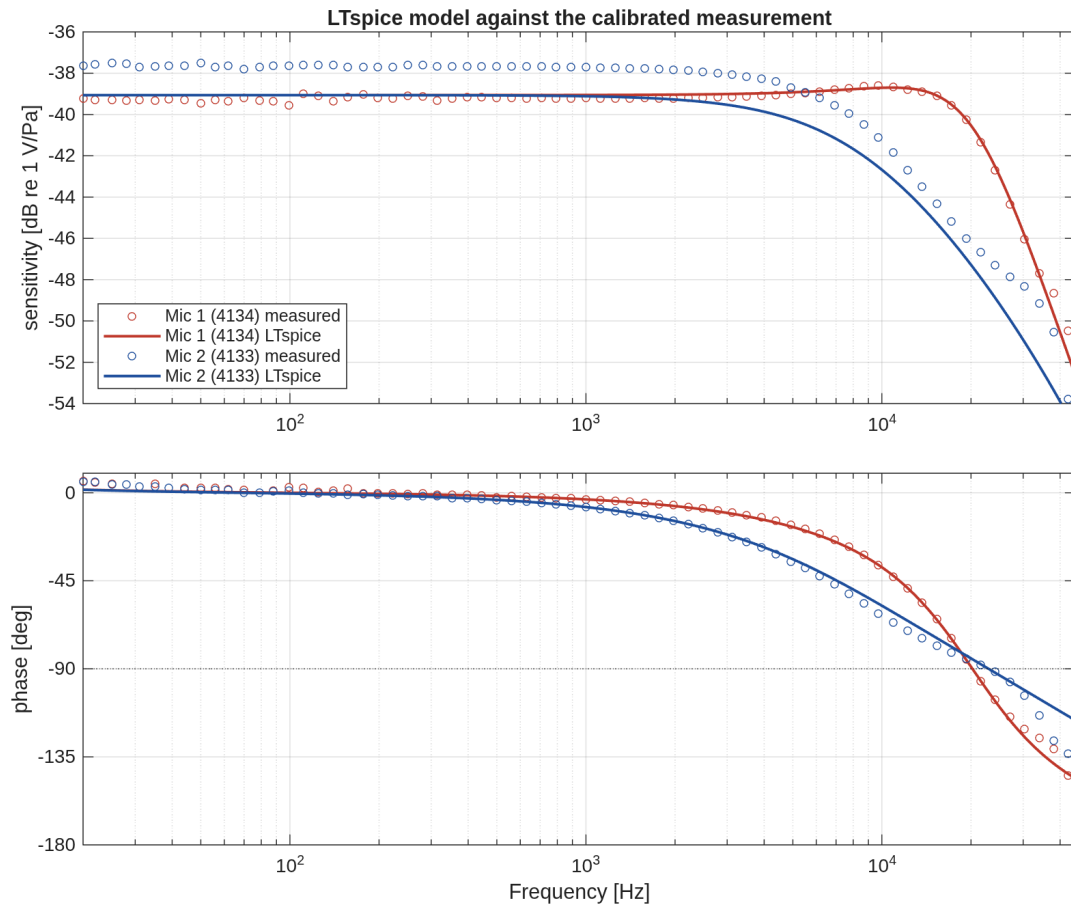


Figure 6: LTspice output (lines) against the calibrated measurement (circles), in absolute sensitivity. The output current source carries the sign of the ground convention, so no extra 180° shift is needed.

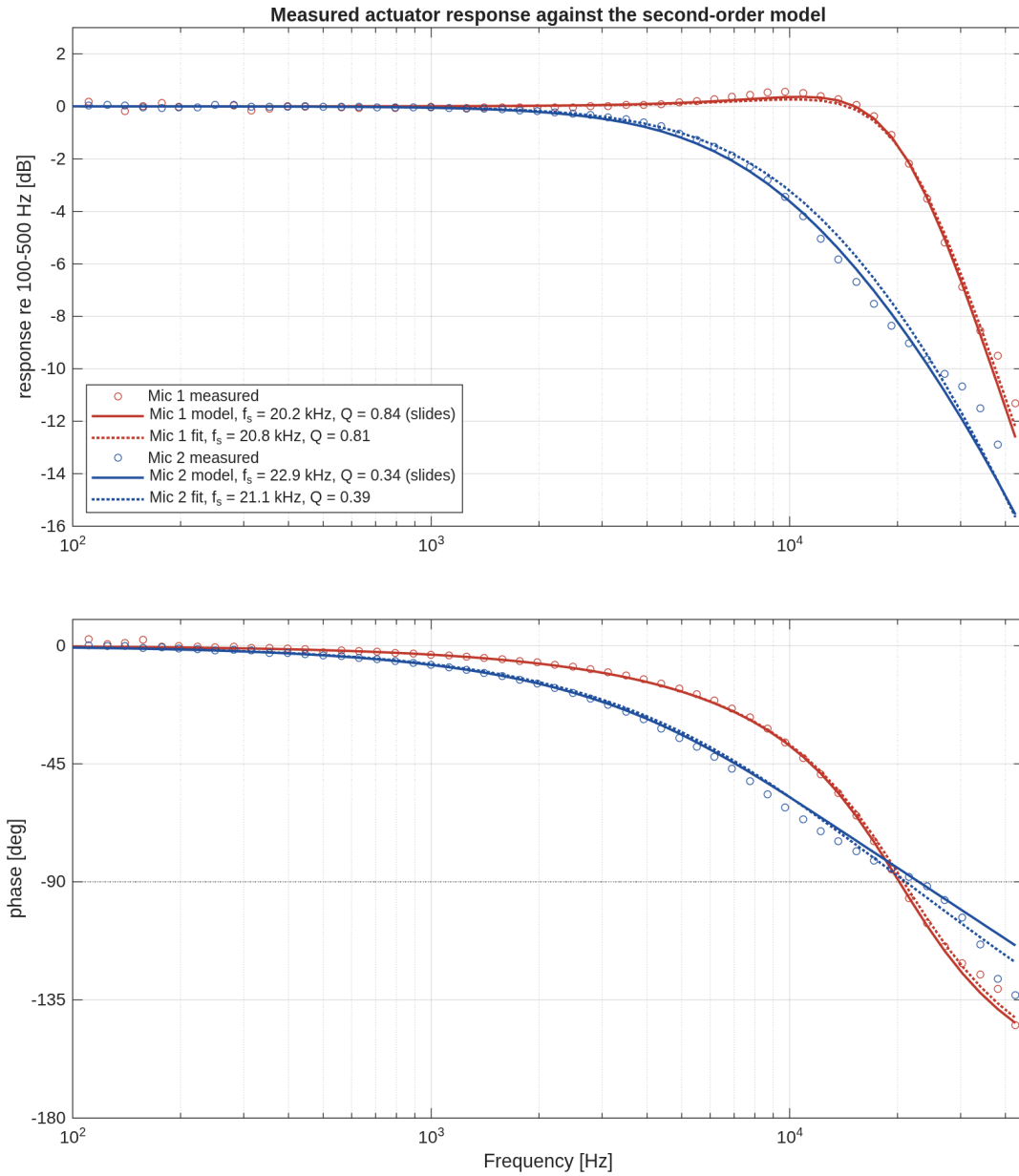


Figure 7: Shape only: measurement normalised to the 100 Hz–500 Hz level, with the model from the slide values (solid) and from the fit (dotted).

- **4134 (Mic 1):** the model follows the measurement within about 1.3 dB and 8° over the whole band, and within a few tenths of a dB up to 20 kHz. The absolute level is only 0.13 dB below the calibration. The lumped model describes this capsule well without any adjustment.
- **4133 (Mic 2):** most of the visible gap in the absolute plot is level, not shape. The model's sensitivity is 1.36 dB below the calibration, because it uses one fixed $EC_{MT}SD/x_0$ for both capsules while the real 4133 and 4134 differ in compliance and gap (the brief's appendix). Aligned at 1 kHz, the shape agrees within 0.5 dB and 5° up to 20 kHz. Above resonance the measurement departs by up to 1.6 dB and 19° : the phase falls slowly between about 12 and 25 kHz and then faster than the model. One lumped R_{AS} and M_{AS} cannot follow the frequency-dependent air-film damping of a free-field capsule everywhere, which is also why the slide reading and the fit differ by 9% in f_s .

7 Conclusion

- At 1 kHz, Mic 1 (B&K 4134) has $M = 10.98$ mV/Pa (-39.2 dB re 1 V/Pa) and Mic 2 (B&K 4133) has $M = 13.03$ mV/Pa (-37.7 dB). The expanded uncertainty is about 2.8% ($k = 2$), dominated by the calibrator.

- The three calibrators agree within 0.2 %, but the 250 Hz method reads 4 % low for both microphones. This is a systematic effect of the method, a clear example of repeatability not implying accuracy.
- The flat actuator response identifies Mic 1 as the pressure-field 4134 ($f_s = 20.2$ kHz, $Q = 0.84$). The strongly damped response identifies Mic 2 as the free-field 4133 ($f_s = 22.9$ kHz, $Q = 0.34$).
- The LTspice model with M_{AS} and R_{AS} from (f_s, Q) reproduces the 4134 closely. The 4133 is reproduced up to its resonance; above it the lumped model reaches its limit, because the capsule's damping is not a single lumped element.
- The actuator gives the shape of the pressure response quickly and with a large signal-to-noise ratio, but not its absolute level and not the free-field response. An acoustic calibration is still needed for the level, and a free-field correction for use in a sound field.

Processing record

Raw data: LabC/rawdata/ (lab-PC folder, unchanged: `datas_part1.mat`, `datas_part2.mat`, `datas_part3.mat`, `all.mat` and the scripts as run). Processing: LabC/matlab/part1_system_reference.m to part4_model.m (run_all.m). Model: LabC/ltspice/gen_labC_ltspice.py--export--verify (LTspice agrees with the closed-form second-order model within 10^{-6}). Brief: DTU 34870, *Lab C – Microphone calibration – E26*.